When Credit AI Is Wrong: Error Costs, Explainability, and Prudential Governance

Israel Cedillo-Lazcano
Department of Law
Universidad de las Américas Puebla, Mexico

Edwin Montes-Orozco
Department of Computing, Electronics and Mechatronics
Universidad de las Américas Puebla, Mexico

October 2026

Abstract

Artificial intelligence in lending creates a supervisory problem when predictive performance, lending-policy economics, and decision explanations are assessed as interchangeable evidence. We develop a framework that separates calibrated credit risk, individualized approval thresholds, explanation stability, and scarce human-review capacity. A historical South German Credit pilot uses 1,000 records, four disjoint fitting, calibration, policy-selection, and test partitions, and 45 fitted models across three model families and three weighting regimes. On the 200-record test set, instance-dependent random forests achieve the highest mean AUROC (0.763), whereas conventional random forests with individualized economic thresholds achieve the lowest mean scenario loss (-0.0146). All economic-policy loss comparisons with the logistic baseline have conditional 95% intervals spanning zero. Explanation stability changes in different directions across model families. Under assumed imperfect review, routing by proximity to the decision threshold outperforms explanation-instability routing at 10% capacity in point estimates. An analytical decomposition identifies the economic determinants omitted by score-only explanations and the costs and benefits of review. A Basel-grounded evidence framework connects these diagnostics to prudential documentation while distinguishing them from applicant-notification requirements. The contribution concerns institution-level model risk and supervisory auditability; the pilot does not estimate systemic spillovers, capital requirements, or observed monetary returns. Financial-stability implications are framed as transmission mechanisms requiring multi-institution validation. Costs and reviewer behavior are assumptions, and external administrative and dossier-level legal validation remain future work.

Keywords: credit-model risk; cost-sensitive learning; explainability; prudential supervision; human review.

JEL classification: G21; G28; C45.

1 Introduction

The use of artificial intelligence in credit allocation raises a financial-supervision question: what evidence is needed to determine whether a lending system is economically sound, stable under model updates, and open to effective review? The Financial Stability Board identifies model risk, data quality, and governance among AI-related vulnerabilities with potential systemic relevance

(Financial Stability Board, 2024). Our study examines a measurable part of that concern at the institution level: the chain from a credit-risk score to an approval policy, its explanation, and a resource-constrained review decision.

A probability of bad credit is not a lending decision. A model with good ranking performance can support an unfavorable policy when the losses from granting a bad loan and the margins from granting a good loan differ across applicants. An economically attractive policy can also produce unstable explanations or change access to credit. Risk management therefore needs to observe the fitted model, calibration, policy parameters, and review process separately. In particular, feature attributions explain a score relative to a reference construction; they do not necessarily explain why an institution adopted the threshold that determined its action.

The prudential and applicant-facing dimensions are related but distinct. The Basel Framework permits an internal ratings-based approach subject to supervisory approval; a benchmark classifier does not automatically qualify as an approved rating system (Basel Committee on Banking Supervision, 2025, CRE20.2). U.S. Regulation B concerns specific principal reasons for adverse action and their correspondence to factors actually considered (Consumer Financial Protection Bureau, n.d.b,n). We use these sources to structure decision documentation, without applying U.S. notification rules retrospectively to historical German loans or treating Basel standards as universally self-executing domestic law.

We study four questions. **RQ1:** How do weighted fitting and individualized thresholds separately affect scenario loss and approval rates? **RQ2:** How stable are explanations and decisions under training resampling? **RQ3:** Under explicit reviewer-error assumptions, how should limited review capacity be routed? **RQ4:** What decision evidence links these diagnostics to prudential oversight, and what additional observations would be required for a financial-stability claim?

Our pilot uses 1,000 historical South German Credit records, with 200 held out for testing. It reports mixed findings: the configuration with the highest AUROC is not the one with the lowest scenario loss; all economic-policy comparisons with the logistic baseline remain statistically inconclusive under conditional bootstrap intervals; and cost-sensitive weighting does not change explanation stability uniformly. Review value also depends on routing and on reviewer behavior. These results support a joint evaluation protocol rather than a claim that one model or governance intervention is universally superior.

The contribution is an audit-oriented framework for credit-model risk: an explicit policy decomposition, a review-value accounting identity, and a source-to-evidence governance rubric. Its financial-stability relevance is a research hypothesis about the consequences of deploying similar policies across institutions, not an estimated macroprudential effect. The present data do not identify cross-bank dependence, observed monetary losses, or regulatory capital requirements. Administrative reason alignment and case-level legal review are specified as external-validation extensions in the appendices.

2 Related Work and Contribution

Cost-sensitive learning distinguishes statistical errors from their consequences. Ballegeer et al. evaluate SHAP and LIME stability for instance-dependent credit classifiers and report reduced explanation stability in their experiments (Ballegeer et al., 2025). Our pilot tests whether this pattern is visible under a different, specified modeling protocol; it does not presume that the effect must hold.

SHAP provides additive feature attributions relative to a specified reference construction (Lundberg and Lee, 2017). LIME fits a local surrogate around an observation (Ribeiro et al., 2016). Their

explanatory objects and neighborhoods differ. Agreement can be useful diagnostically, but disagreement does not identify which method is legally adequate. Nor does exact additive reconstruction establish causal faithfulness or a realistic counterfactual.

Barocas, Selbst, and Raghavan analyze assumptions behind feature-highlighting explanations and principal reasons, including the difficulty of translating changes in model features into meaningful actions (Barocas et al., 2020). We operationalize a related evidentiary distinction: a score factor may be faithfully computed but insufficient to account for an institution's policy or unsupported by a public record.

The financial-governance contribution differs from a new prediction algorithm: we connect model validation to the approval rule and the allocation of review resources. The FSB's assessment motivates governance and monitoring of AI-related vulnerabilities, but does not validate any particular credit-scoring model (Financial Stability Board, 2024). Our analysis supplies institution-level diagnostics for this agenda; it does not measure market correlations or systemic amplification. We do not introduce a new classifier or infer legal sufficiency from technical metrics.

3 Data and Scope

3.1 Executed historical-credit pilot

We use the corrected South German Credit dataset, UCI 522 (UCI Machine Learning Repository, 2019). Its 1,000 records comprise 700 good and 300 bad credits. We code $Y = 1$ for bad credit, which is broader than a documented modern default event. The historical sample oversamples bad credits. Consequently, fitted probabilities and policy estimates describe this benchmark distribution; they are not calibrated deployment default probabilities.

The original amount variable underwent an unknown monotonic transformation. We therefore construct a scenario exposure index, not a monetary EAD estimate. Sex cannot be cleanly recovered from the combined personal-status/sex coding. We exclude that variable, age, and foreign-worker status from fitting; we retain age only for a limited audit. Excluding these variables does not eliminate proxies. The remaining 17 fields contain two numerical predictors (amount and duration); other codes are one-hot encoded to avoid imposing spurious linear ordering.

A fixed stratified split contains 550 fitting records, 150 calibration records, 100 policy-selection records, and 200 test records. Preprocessing is fitted on the fitting partition only. Exact split indices and the data digest are required to reproduce the reported fitted models; they are not included in the present submission package. Five bootstrap resamples of the same fitting partition, seeded 11, 22, 33, 44, and 55, are paired across configurations. This evaluates training-resample sensitivity, not pure optimizer randomness or independent test replications.

3.2 Planned external validation arms

The original research design's LendingClub arm remains a planned modern-credit extension. It requires a documented accepted-loan extract, follow-up-aware chronological splits, and exclusion of post-origination fields. Funded-loan selection and unresolved outcomes must be reported. Rates, grades, repayments, collections, and recoveries must not silently leak into a pre-decision risk model. No LendingClub experiment is reported here.

The separate HMDA arm studies historical decisions and reported denial categories, not repayment (Appendix A). Its public dictionary distinguishes originations, approvals not accepted, denials, and other application outcomes (FFIEC and Consumer Financial Protection Bureau, n.d.). The proposed adapter would filter a homogeneous 2025 conventional home-purchase, first-lien, owner-occupied, site-built one-to-four-unit cohort; it excludes preapproval requests and actions outside

1–3. Training and test institutions are disjoint. Data acquisition, extract version, and institution composition must be frozen before interpreting results. No common frontier will combine costs from one population with disparities from another.

4 Decision Model

4.1 Economic policy and evaluation loss

Let A denote approval and R rejection. Relative to not granting a loan, the scenario loss is

	$Y=0$	$Y=1$
A	$-\Pi_i$	L_i
R	0	0

(1)

where $\Pi_i > 0$ is performing-loan margin and $L_i > 0$ the loss of a bad loan. Rejection does not incur an additional foregone-margin charge: doing so would double-count the difference between actions. Rights and applicant harm are not monetized by this matrix.

For calibrated risk p_i, expected approval loss is

$$g_i = p_i L_i - (1 - p_i)\Pi_i. \tag{2}$$

The economic policy approves exactly when

$$p_i < \tau_i = \frac{\Pi_i}{L_i + \Pi_i}. \tag{3}$$

The test statistic is the average realized-label *scenario* loss $C_i(a_i, Y_i)$, not actual cash flow. Model-expected loss and realized-label scenario loss are different quantities and must be recorded separately in a replication. Negative loss means positive scenario net return. We report differences instead of percentage savings because a baseline can be negative or close to zero.

For the pilot, with transformed amount q_i, fitting median $q_{.5}$, and duration d_i in months, we set

$$e_i = q_i / q_{.5}, \tag{4}$$

$$\Pi_i = e_i[0.06 + 0.003 \min(d_i, 60)], \tag{5}$$

$$L_i = 0.60 e_i + 0.05. \tag{6}$$

All values are normalized units over the loan horizon. The coefficients are experimental assumptions. Sensitivity varies the loss coefficient over $\{0.3, 0.6, 0.9\}$ and multiplies margin by $\{0.5, 1, 1.5\}$. Historical repayment timing, funding costs, and realized recoveries are not available to validate these scenarios.

4.2 Learning, calibration, and thresholds

Logistic regression (LR), random forest (RF), and gradient boosting (GB) are fitted under standard, balanced, and instance-dependent cost-sensitive (IDCS) weights. IDCS weights are

$$w_i = Y_i L_i + (1 - Y_i)\Pi_i, \tag{7}$$

normalized to mean one. This is a weighted surrogate objective, not an exact optimizer of realized lending profit. Balanced weights give each observed class equal total weight. Hyperparameters are fixed, with no test-driven tuning: LR $C = 1$; RF 120 trees, minimum leaf size 8; GB 100 depth-2

trees, learning rate 0.05, minimum leaf size 10.

Weighted outputs are not assumed to be probabilities. An unweighted logistic calibration maps raw scores to probabilities using only the 150-record calibration set. Scores are LR/GB margins and RF probabilities. The U policy selects a common threshold on a fixed grid from 0 to 1 in increments of 0.005, minimizing policy-partition scenario loss; ties favor the smallest threshold. The E policy uses τ_i directly. Thus even the U baseline is financially tuned; differences do not merely compare E to an arbitrary 0.5 threshold.

4.3 Risk explanations versus policy determinants

Grouped SHAP values explain the calibrated *logit*, not the calibrated probability. For LR, linear interventional attributions use fitting-set means. For RF/GB, TreeSHAP uses the fitted tree-path reference. The affine calibration slope rescales the raw attributions. One-hot attributions are summed into their original variables. Because reference constructions differ, cross-family rankings should be interpreted cautiously; within-family comparisons are the main stability diagnostic.

We verify additive reconstruction numerically before calibration. This checks implementation consistency only. It is not a held-out surrogate-fidelity test, does not establish plausible perturbations, and does not make the attributions causal. For each record we select up to three positive grouped contributions. Empty sets receive an undefined Jaccard score, not perfect agreement. Jaccard overlap is reported for equal-decision pairs and for joint denials, alongside the pairwise decision-flip rate. Pair counts are not independent sample sizes.

A risk explanation alone omits the economic threshold. For strictly positive costs, the same decision can be written

$$D_i = \operatorname{logit}(p_i) + \log L_i - \log \Pi_i, \tag{8}$$

with rejection when $D_i \geq 0$. This identity identifies a simple auditable policy decomposition: the score and the two economic terms. Its latter terms are policy components, not additional independent borrower-feature SHAP values. Since amount and duration can affect both risk and policy, treating all these components as independent causal effects would be incorrect. Case dossiers therefore retain p_i, τ_i, Π_i, L_i, g_i, and ranked score factors separately. The pilot does not claim a joint SHAP decomposition of the full policy.

LIME is evaluated on 20 fixed random test records for the seed-11 standard RF, using three explainer seeds and 1,500 neighborhood samples. Numerical features are continuous and coded categorical features are treated categorically. The reported weighted R^2 is an in-neighborhood fit diagnostic, not out-of-neighborhood validation. Positive LIME coefficients and positive SHAP contributions are not identical explanatory objects; their top-three overlap is descriptive only. LIME's marginal perturbations can violate joint feature dependencies.

5 Human Review and Statistical Evaluation

Review routing uses a fixed standard-RF ensemble, chosen as a designated scenario model rather than by its test ranking. Its probability is the mean of five calibrated probabilities. Uncertainty is the percentile rank of closeness to τ_i; instability is one minus average pairwise reason overlap. The combined score weights them equally. All routing scores are computable without test outcomes. Budgets are 5%, 10%, and 20%.

Reviewer behavior is hypothetical. Let $b_i = 1$ when the automated action disagrees with the full-information oracle (approve good, reject bad). If review corrects a fraction c of wrong actions

and introduces an error in a fraction r of correct actions, its expected wrong-action probability is

$$v_i = b_i(1-c) + (1-b_i)r. \tag{9}$$

Expected reviewed loss is

$$\ell_i^H = -\Pi_i(1-Y_i) + v_i[Y_i L_i + (1-Y_i)\Pi_i] + h. \tag{10}$$

We evaluate $(c, r) = (0, 0)$, $(0.5, 0.05)$, and $(1, 0)$, with $h = 0.01$. The last scenario is an oracle bound, not human performance. Labels enter retrospective evaluation only. Random routing is integrated analytically over uniform selection. Review cost is charged even when no correction occurs.

We report AUROC, average precision, Brier score, approval rate, and scenario loss. Predictive metrics describe the benchmark's selected class distribution. Differences in loss use 2,000 paired bootstrap resamples of test records, after averaging per-record loss over training seeds. Intervals are conditional on one split and fixed modeling choices. They do not include population shift, uncertainty in cost assumptions, or repeated-test-set variation; comparisons are exploratory and not multiplicity adjusted.

5.1 Review-value accounting

Define $W_i = Y_i L_i + (1-Y_i)\Pi_i$. Before review, the realized-label scenario loss is $\ell_i^A = -\Pi_i(1-Y_i) + b_i W_i$. Subtracting the reviewed loss gives

$$B_i = \ell_i^A - \ell_i^H = \big[cb_i - r(1-b_i)\big]W_i - h. \tag{11}$$

This is an algebraic consequence of the stated reviewer model, not an additional experiment. It shows that review creates value by correcting costly errors, can destroy value by overturning correct decisions, and consumes resources regardless of outcome. Explanation instability is not a sufficient statistic for any of these terms.

For a prospective workflow, taking conditional expectations under the same constant-c, constant-r assumptions gives the expected review benefit

$$\widehat{B}_i = (c+r)\mathbb{E}[b_i W_i \mid x_i] - r\mathbb{E}[W_i \mid x_i] - h. \tag{12}$$

The joint expectation matters because error probability and economic stakes need not be independent. Estimating these terms would require validated reviewer data and deployment-relevant outcomes. The pilot evaluates routing proxies; it does not implement or establish the superiority of this prospective policy.

6 Pilot Results

6.1 Prediction and policy outcomes

Table 1 separates model fitting from policy effects. RF-IDCS has the largest mean AUROC (0.763) and lowest Brier score (0.173) among the nine fitted configurations. However, RF-standard-E has the lowest mean scenario loss (-0.0146), compared with -0.0121 for RF-IDCS-E. The highest AUROC therefore does not identify the lowest-loss configuration in this pilot.

RF-standard's approval rate changes from 38.3% under U to 29.3% under E. For RF-IDCS, it changes from 33.0% to 37.5%. Economic thresholds do not uniformly reduce access: the effect

Table 1: Held-out performance: means over five paired training bootstraps. U: validation-selected uniform threshold; E: individualized economic threshold. Loss is a scenario proxy; lower is better and negative means positive net return. SD measures training variability on one fixed test set, not population uncertainty.

Model	Training	Policy	AUROC	AP	Brier	Approve (%)	Loss	Loss SD
GB	IDCS	E	0.751	0.574	0.175	32.4	-0.0068	0.0158
GB	IDCS	U	0.751	0.574	0.175	24.6	-0.0074	0.0093
GB	balanced	E	0.752	0.562	0.176	28.7	-0.0106	0.0069
GB	balanced	U	0.752	0.562	0.176	41.6	-0.0013	0.0168
GB	standard	E	0.744	0.550	0.179	24.5	-0.0091	0.0051
GB	standard	U	0.744	0.550	0.179	31.4	-0.0099	0.0065
LR	IDCS	E	0.731	0.537	0.182	26.9	-0.0079	0.0043
LR	IDCS	U	0.731	0.537	0.182	28.9	-0.0018	0.0067
LR	balanced	E	0.744	0.537	0.178	25.2	-0.0096	0.0068
LR	balanced	U	0.744	0.537	0.178	31.9	-0.0119	0.0080
LR	standard	E	0.742	0.536	0.179	26.1	-0.0080	0.0074
LR	standard	U	0.742	0.536	0.179	32.5	-0.0080	0.0068
RF	IDCS	E	0.763	0.564	0.173	37.5	-0.0121	0.0143
RF	IDCS	U	0.763	0.564	0.173	33.0	-0.0095	0.0093
RF	balanced	E	0.757	0.552	0.176	31.3	-0.0130	0.0054
RF	balanced	U	0.757	0.552	0.176	48.9	-0.0029	0.0118
RF	standard	E	0.759	0.550	0.176	29.3	-0.0146	0.0035
RF	standard	U	0.759	0.550	0.176	38.3	-0.0089	0.0057

depends on the fitted score and the common threshold selected on the policy partition. For standard LR, the two policies yield almost identical mean losses, -0.00804 and -0.00799.

Table 2: Paired change in loss versus LR-standard-U. Percentile 95% bootstrap intervals resample the same 200 test records, after averaging losses over training seeds. All E comparisons include zero.

Model	Training	Δ loss	95% interval
GB	IDCS	0.0012	[-0.0204, 0.0282]
GB	balanced	-0.0026	[-0.0168, 0.0141]
GB	standard	-0.0011	[-0.0145, 0.0138]
LR	IDCS	0.0001	[-0.0111, 0.0125]
LR	balanced	-0.0015	[-0.0099, 0.0056]
LR	standard	0.0000	[-0.0066, 0.0066]
RF	IDCS	-0.0041	[-0.0193, 0.0150]
RF	balanced	-0.0050	[-0.0155, 0.0079]
RF	standard	-0.0065	[-0.0201, 0.0088]

The mean RF-standard-E difference from LR-standard-U is -0.00653 with a conditional 95% interval of $[-0.02009, 0.00877]$. Table 2 shows that every economic-policy comparison against this baseline includes zero. We therefore describe point estimates and do not claim established superiority. Training variation is also material: RF-IDCS-E loss SD is 0.0143, compared with 0.0035 for RF-standard-E.

6.2 Explanation stability is model dependent

Table 3 does not support a universal reduction of stability under IDCS. Among joint denials, RF overlap increases from 0.455 to 0.522; LR increases from 0.440 to 0.473; GB decreases from 0.445 to 0.434. The sets of jointly denied cases differ between regimes, so these differences are descriptive rather than matched causal effects of weighting. The pairwise flip rate ranges from 0.146 to 0.210. Reporting overlap alone would hide this action instability.

Figure 1 shows the empirical configuration space. A low-loss point need not have the highest

Table 3: Grouped positive top-3 SHAP overlap across training bootstraps. Same: same-action pairs; Denied: joint denials. Flip: pairwise decision disagreement under E. These are descriptive means, not causal or legal validity measures.

Model	Training	Same	Denied	Flip
LR	standard	0.475	0.440	0.150
LR	balanced	0.453	0.426	0.151
LR	IDCS	0.481	0.473	0.210
RF	standard	0.455	0.455	0.153
RF	balanced	0.460	0.464	0.146
RF	IDCS	0.483	0.522	0.148
GB	standard	0.442	0.445	0.196
GB	balanced	0.451	0.464	0.195
GB	IDCS	0.409	0.434	0.204

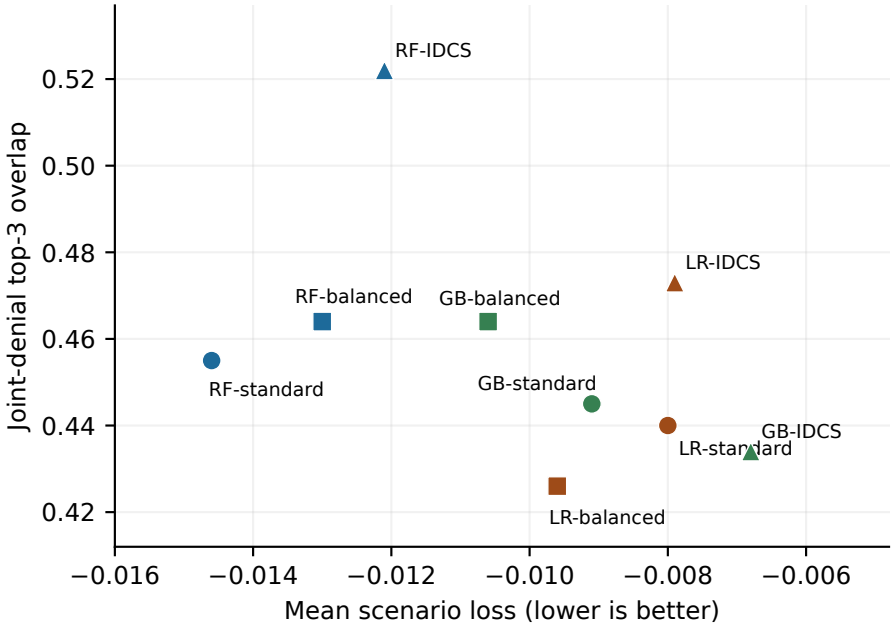


Figure 1: Observed cost–stability configurations under E. Lower loss and higher denial overlap are preferred within this scenario. Points are reconstructed from the rounded summaries in Tables 1 and 3, and are not a universal deployment frontier; overlap uses configuration-specific joint-denial subsets.

overlap, but nine configurations on one historical split do not establish a structural trade-off. The result narrows the claim relative to the initial hypothesis and motivates larger matched-denial analyses.

The mean weighted in-neighborhood fit R^2 was 0.491; LIME explainer-seed top-three overlap was 0.783. The mean cross-method overlap was 0.522.

These numbers distinguish retraining instability from explainer randomness. They should not be compared as if they measured the same disturbance. In particular, modest local fit and cross-method disagreement argue against turning an unreviewed top-three list directly into an adverse-action notice.

6.3 Review value depends on routing and reviewer errors

The RF-standard ensemble's unreviewed loss is −0.01266. This differs from the mean individual-seed loss in Table 1 because thresholding the mean probability need not equal averaging thresholded actions. At 10% review capacity under the imperfect-review scenario, uncertainty routing changes loss by −0.00841, compared with −0.00364 for combined routing, −0.00281 for random routing,

Table 4: Change in mean scenario loss after imperfect review (50% error correction; 5% error introduction; cost 0.01 per reviewed case). Negative changes improve the RF-standard ensemble baseline. Random routing is its exact selection expectation.

Routing	5%	10%	20%
random	-0.0014	-0.0028	-0.0056
uncertainty	-0.0027	-0.0084	-0.0080
instability	0.0008	-0.0005	-0.0016
combined	-0.0014	-0.0036	-0.0083

and −0.00046 for instability-only routing. At 5%, instability-only review increases loss by 0.00079.

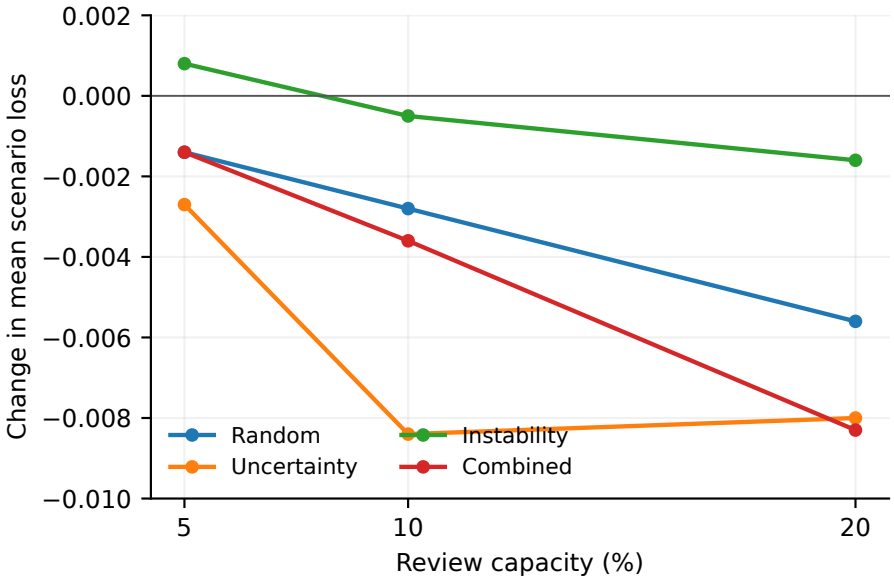


Figure 2: Imperfect-review scenario. Negative changes reduce loss relative to automation. Lines are reconstructed from Table 4 and describe hypothetical reviewer behavior, not observed analyst performance.

The result is not that unstable explanations should be ignored. Rather, explanation instability and expected correction value are different quantities. A case can have unstable score factors while being a correct automated action; review may then introduce an error. With no reviewer improvement, every strategy adds exactly 0.01 times the budget to mean loss. This provides a useful accounting check and prevents review from being treated as cost-free abstention.

6.4 Sensitivity and subgroup audit

Table 5: Sensitivity of the fixed RF-standard ensemble to loss and margin scenarios. Models are not retrained. The loss coefficient is a scenario coefficient applied to a transformed exposure index, not an estimated recovery rate.

Loss coeff.	Margin mult.	Approve (%)	Loss
0.3	0.5	24.5	-0.0100
0.3	1.0	51.5	-0.0350
0.3	1.5	67.5	-0.0781
0.6	0.5	6.5	-0.0016
0.6	1.0	28.5	-0.0127
0.6	1.5	42.5	-0.0382
0.9	0.5	1.5	-0.0014
0.9	1.0	14.0	-0.0143
0.9	1.5	28.5	-0.0197

Table 5 holds the fitted ensemble fixed while changing policy economics. These are scenario evaluations, not additional independently fitted experiments. Conclusions about financial superiority must be conditional on the selected loss and margin assumptions.

Table 6: Exploratory age audit of the RF-standard ensemble, E policy. BA: approval among bad credits; GR: rejection among good credits. Group denominators are small (bad: 12/48; good: 21/119). No unlawful discrimination inference is made.

Age	n	Approve (%)	BA (%)	GR (%)
age <25	33	9.1	0.0	85.7
age ≥25	167	32.3	12.5	59.7

The age audit in Table 6 concerns only the standard-RF ensemble. Applicants younger than 25 have an approval rate of 9.1%, versus 32.3% for the older group. There are only 33 younger test records, including 12 bad credits. Zero observed bad-credit approvals in that subgroup does not establish zero population error. No inferential claim or legal conclusion follows from these descriptive rates; institutional context, selection, legitimate decision factors, and adequate samples are missing.

7 Prudential Governance and Decision Documentation

7.1 Distinct regulatory purposes

A technical factor becomes a candidate notice reason only after its direction, materiality, factual support, and correspondence to the actual decision process have been established. Public HMDA alignment cannot itself establish these properties. Legal review should inspect the recorded decision rule, feature provenance, calibrated output, economic terms, and available notice evidence together.

International standards and domestic application. The Basel Framework supplies an internationally influential prudential reference. The Basel Committee has no supranational authority and its decisions have no independent legal force; application depends on implementation by the relevant jurisdiction (Basel Committee on Banking Supervision, n.d., Secs. 3, 5). Accordingly, each dossier must identify its jurisdiction, decision date, institution, applicable domestic rule, capital approach, and exposure class before a Basel provision is assigned direct relevance. Regulation B addresses U.S. adverse-action notifications; Basel addresses prudential risk and capital; DORA addresses operational resilience (European Parliament and Council of the European Union, 2022, Arts. 2, 5–6). A record satisfying one purpose does not establish sufficiency for the others.

CRE20.4 motivates counterparty-specific due diligence within its stated scope. CRE20.63, CRE20.65, and CRE20.70 distinguish retail eligibility and real-estate categories rather than prescribing notice wording (Basel Committee on Banking Supervision, 2025). CRE36.10 concerns documented assignment among rating systems. The receivables provisions are narrower: CRE36.117 concerns purchased-receivables monitoring, while CRE36.133–CRE36.140 concern eligible financial receivables, legal certainty, and risk management (Basel Committee on Banking Supervision, 2022). We retain those links only where the transaction and prudential treatment make them applicable. The historical pilot and public HMDA records do not establish that applicability.

Legal contribution and evidence criteria. Our legal analysis links five criteria—specificity, factual support, decision correspondence, material completeness, and comprehensibility—to inspectable decision evidence. Appendix B, Table 8, retains the legal coauthor's criterion-level scores

(2, 2, 1, 2, and 1) and regulatory justifications. These are normative assessments of the criteria, not loan-level validation results or a numerical scale prescribed by Basel. The rubric supplies a structured documentation layer for supervision while preserving the distinction between prudential records and applicant-facing reasons.

Interpretation and applicability. The table reports the legal coauthor's justification rather than treating every cited section as a universal lending obligation. In particular, CRE20.4 supplies the explicit due-diligence link to each counterparty's circumstances, while CRE20.63/20.65/20.70 address exposure classification. The CRE36 links retain their receivables and rating-system scope. The comprehensibility score reflects the legal coauthor's reasoning about information available to the user and collateral; actual notice clarity must still be assessed using the relevant document and notification rules (Consumer Financial Protection Bureau, n.d.b,n).

Dossier assessment procedure. A qualified legal-domain expert will assess a prespecified stratified sample using these five criteria. For the subsequent dossier study, the same numerical range is applied to inspectable case evidence: 0 means that inspectable evidence fails to support the criterion; 1 means partial support with a material gap; 2 means adequate support for the specified assessment purpose. *Not evaluable* means that the necessary evidence is inaccessible or the assessment cannot be made. The reviewer records evidence locators, applicable rules, a short justification, and information needed to resolve gaps. A Basel provision outside the dossier's scope is marked not applicable as a source link, while the underlying criterion can still be assessed under other applicable sources.

For example, a documented high DTI may support a specific adverse reason, but only if its provenance and actual role in the denial are established. A valid collateral document may support a prudential record without explaining whether a cost threshold or override determined the action. The supplied criterion-level scores do not assign scores to either example.

No dossier-level ratings have been collected. The planned single-expert study will not estimate inter-rater reliability and will be interpreted as structured expert review rather than consensus validation. Non-evaluable cases must not be imputed as successes, and population prevalence cannot be inferred from unweighted stratified dossiers. Reviewer qualifications, sampling, ethics requirements, and the frozen coding manual will be documented before the empirical review.

8 Supervisory Implications and Financial-Stability Scope

8.1 What an institution-level audit should retain

The pilot supports a concrete supervisory question: can a reviewer reconstruct the policy action from the risk score, calibration version, economic assumptions, and any override? A record should keep the applicant inputs and provenance, model and calibration identifiers, p_i, τ_i, L_i, Π_i, the action, the explanation reference construction, and the review outcome. This is our proposed evidence specification, not a claim that every item is legally mandated in every jurisdiction.

Three findings guide monitoring. First, validation should report loss and approval changes alongside discrimination and calibration: the observed AUROC ranking did not select the lowest-loss policy. Second, action flips and explanation overlap are complementary measures: stable reasons within common denials can coexist with changes in who is denied. Third, review should be evaluated after its resource cost and possible introduced errors. Equation 11 explains why directing all review toward unstable explanations can be economically ineffective.

Basel's IRB validation provisions concern validation of rating systems and risk estimates, com-

parisons with realized outcomes, relevant external data, and economic-cycle variation (Basel Committee on Banking Supervision, 2022, CRE36.122–CRE36.126). These provisions provide a more direct prudential link to model validation than applicant-notice requirements, within their specified scope. Our benchmark supplies neither a complete credit cycle nor an approved IRB system. It illustrates diagnostics that would need to be embedded in a substantially richer institutional validation process.

8.2 Portfolio aggregation is not systemic-risk estimation

For a fixed book of candidate loans and actions a_i, the expected scenario loss is

$$\mathbb{E}[C_{\mathcal{P}} \mid x, a] = \sum_{i \in \mathcal{P}} \mathbf{1}\{a_i = A\}\big[p_i L_i - (1 - p_i)\Pi_i\big]. \tag{13}$$

Linearity of expectation does not require independent defaults, but it does require valid marginal probabilities and economically meaningful loss inputs. The pilot probabilities reflect a selected historical benchmark, and the exposure index is not monetary exposure at default. Equation 13 is therefore an accounting bridge, not a reported estimate of bank losses.

Tail risk additionally depends on joint outcomes:

$$\mathrm{Var}(C_{\mathcal{P}}) = \sum_i \mathrm{Var}(C_i) + 2\sum_{i<j} \mathrm{Cov}(C_i, C_j). \tag{14}$$

The present experiment does not estimate these covariances, common exposures, funding constraints, or spillovers. A lower mean scenario loss in Table 1 consequently cannot establish resilience to aggregate shocks.

8.3 Testable transmission mechanisms

As a research hypothesis, common models and similar threshold changes could synchronize credit expansion or contraction across institutions. Shared model errors could also accumulate exposures that appear individually acceptable, while review-capacity limits could delay correction. These are proposed mechanisms, not effects identified by this sample. Testing them requires lender identifiers, common-exposure information, repeated decisions through time, deployment-level calibration, and reviewer outcomes. Financial-stability relevance thus comes from linking decision governance to these measurable extensions; explanation quality alone is insufficient evidence of systemic-risk mitigation.

9 Limitations and Next Experiments

External validity. A small historical credit sample with deliberately altered class prevalence cannot demonstrate modern lending performance. The amount transformation prevents observed monetary-loss claims. A single test split cannot support deployment guarantees. Modern accepted-loan data still require attention to funded-loan selection, maturity, and temporal drift.

Explanatory validity. Tree-path and linear reference constructions differ. Grouping preserves additive sums but does not make grouped one-hot values equal to a conditional group-Shapley game. Positive contributions are relative to references, not necessarily the institution's principal reasons. Dependence-aware perturbations, held-out local fidelity, common-denial matched comparisons, and explanations of the complete nonlinear policy remain necessary extensions.

Human and legal validity. Review outcomes are analytical scenarios with constant correction and introduction probabilities. Real reviewers may have heterogeneous performance, delays,

anchoring, and unequal effects across groups. HMDA has no repayment labels, and its reporting categories are not original notices. Statistical disparity is not a finding of unlawful discrimination. The legal coauthor's assessment in Table 8 is based on the regulatory sources; a structured dossier-level expert assessment has not yet been performed. Basel provision applicability and domestic implementation must be established separately for each institutional setting. Because the planned assessment relies on a single legal-domain expert, it will not estimate inter-rater reliability or establish consensus across legal reviewers.

Uncertainty and governance. Bootstrap intervals condition on a fixed benchmark. They do not account for unknown policy costs or geographic and temporal transport. Stability or disparity cutoffs would be experimental governance constraints, not legal safe harbors. A larger study should select configurations on validation data, freeze the governance criteria, and evaluate cost, stability, coverage, and review burden on a fresh test cohort.

The next empirical steps are a mature-cohort LendingClub replication, a versioned multi-institution HMDA analysis, and a structured legal-domain expert review of the prespecified dossier sample. These should be published with explicit completion status rather than folded into the pilot as if already observed.

10 Conclusion

Credit-model supervision requires separating predictive risk from policy economics, explanation behavior, and the value of review. In this historical pilot, the best AUROC does not identify the lowest mean scenario loss; the economic-policy comparisons remain uncertain; and explanation stability responds differently across model families. Hypothetical review can improve or worsen outcomes depending on routing, error correction, introduced errors, and cost. The policy and review decompositions make those mechanisms auditable rather than treating explanations or human involvement as guarantees.

Our Basel-grounded legal contribution specifies the evidence needed to inspect the decision process while preserving the distinct purposes of prudential supervision and applicant notification. The resulting framework addresses institution-level model risk relevant to financial regulation. Demonstrating monetary benefits, legal sufficiency, or systemic resilience requires modern multi-institution data and observed review outcomes. The present contribution is a bounded pilot and a supervisory evaluation design, not a financial-stability impact estimate.

Data, code, and study status

South German Credit and its dictionary are publicly available from UCI under CC BY 4.0 (UCI Machine Learning Repository, 2019). This submission package contains the manuscript source, bibliography, and two summary figures. The figures reconstruct the rounded manuscript tables; they do not represent additional experiments. The underlying training code, exact split manifest, predictions, and bootstrap outputs are not part of this submission package, so the reported fitted-model results and intervals cannot be independently reproduced from the package alone. HMDA analysis, LendingClub replication, dossier-level legal ratings, and observed human-review evaluation remain planned validation arms. Criterion-level legal assessments are reported separately in Appendix B.

A Administrative Reason Alignment: External-Validation Protocol

The proposed HMDA adapter is an extension, not an executed result. It would fit a denial surrogate using income, amount, term, property value, loan-to-value ratio, and debt-to-income ratio (DTI). Denial codes, protected attributes, lender identifiers, automated-underwriting outcomes, interest rates, and post-decision fields are excluded as predictors. Public values may be binned, rounded, absent, or exempt; preprocessing does not restore their original precision.

Table 7: Predefined HMDA reason observability. This is a mapping specification, not empirical alignment results. Codes follow the official public dictionary.

Reason code/family	Status	Evaluation rule
1: Debt-to-income ratio	Direct field	Evaluate only when a usable DTI value and reported code 1 coexist; public bins limit detail.
2: Employment history	Unavailable	Income is not employment history; no substitution.
3: Credit history	Unavailable	Score type is not a score or documented credit history.
4: Collateral	Proxy only	LTV/property value do not establish the lender's collateral deficiency; excluded from primary alignment.
5: Insufficient cash	Unavailable	Income is not cash available for down payment/closing.
6–8: Other listed reasons	Unavailable	Unverifiable information, incomplete application, and mortgage-insurance denial need additional evidence.
9: Other	Unevaluable	The public code alone does not identify a specific fact.
10/1111, missing	Not a substantive reason	Not applicable, exempt, and missing values are not evidence that no reason exists.

For a denied record, let D_i be supported reported families and R_{ik} supported families among the top-k positive surrogate factors. We define recall as $|D_i \cap R_{ik}|/|D_i|$ for nonempty D_i. Crucially, ranking precedes filtering: unavailable categories do not promote DTI into the top positions. The initial adapter would support only DTI, making its result a *DTI-specific* recall analysis, not general reason recovery. Precision over a single supported family is potentially uninformative and is not promoted as a headline outcome.

Coverage must accompany recall: the fraction of denied applications and the fraction of reported reasons that are evaluable. Missingness, exemption, and unevaluable categories remain visible. The surrogate describes its own approximation of historical decisions, not the lender's internal model, original notice, or normative creditworthiness. A future analysis should use institution-level bootstrap intervals, group denominators, and a frozen extract before testing transportability.

B Legal Coauthor's Source-to-Evidence Assessment

The following table preserves the supplied criterion-level assessment and its regulatory justifications. Its scores are normative judgments, not observed case-level performance. The source links must be interpreted within the exposure-specific scope discussed in Section 7; they are not universal adverse-action notification rules. A subsequent dossier study will require the sampling and assessment procedure described in the main text.

Table 8: Legal coauthor's assessment of the five source-to-evidence criteria. Criterion-level judgments; the scores do not measure loan-level compliance. The cited Basel provisions are applied within their exposure-specific scope; individual dossier evaluation remains a separate empirical step.

Criterion	Evidence requested	Score	Legal justification
Specificity	The targets must be consistent with banks' overall risk profiles and current operating environments building on Section 20.63 and 20.70 of the Basel Framework. (Basel Committee on Banking Supervision, 2025)	**2**	Based on the macroprudential regulation, the bank has to consider elements of the real economy at the time of the decision to analyze and classify each individual profile. Additionally, it has to consider the retail and real-estate exposures (Section 20.63 and 20.70 of the Basel Framework) for credit risk, considering, for instance, the elements set in Section 20.65 of the Basel Framework
Factual Support	Source field, value, provenance, missingness for the recognition of financial receivables under Section 36.133 of the Basel Framework. (Basel Committee on Banking Supervision, 2022)	**2**	As set out in Sections 36.135, 36.136, 36.137, 36.138, 36.139, and 36.140, banks have to identify the value, the legal reach, the geographical reach, exposures, among other quantifiable elements of the documents involved in a credit transaction.
Decision Correspondence	Actual model, policy, and override record. (Basel Committee on Banking Supervision, 2022)	**1**	Banks under the Basel Framework, and based on their own experiences, can rely on internal models for credit profiling. Certainly, as set in Section 36.10, these models have to be adjusted to different categories of risk, and if they are developed from a common basis, there could be some operational challenges.
Material Completeness	All documented principles and determinants set in Section 36.117 of the Basel Framework. (Basel Committee on Banking Supervision, 2022)	**2**	Banks have the obligation to pair their obligations found in the Sections invoked above with the proper documentation that will allow them, in turn, to make the proper calculation of their capital.
Comprehensibility	Applicant-oriented wording and context	**1**	If we read the Basel Framework, we could be tempted to assign a 0 to this variable. However, despite the natural asymmetries that define this market, the agreements contain the relevant information for the user, particularly if we consider that there could be collateral involved.

References

Matteo Ballegeer, Matthias Bogaert, and Dries F. Benoit. Evaluating the stability of model explanations in instance-dependent cost-sensitive credit scoring. *European Journal of Operational Research*, 326(3): 630–640, 2025. doi: 10.1016/j.ejor.2025.05.039.

Solon Barocas, Andrew D. Selbst, and Manish Raghavan. The hidden assumptions behind counterfactual explanations and principal reasons. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 2020. URL https://arxiv.org/abs/1912.04930.

Basel Committee on Banking Supervision. Basel Framework CRE36: IRB approach—minimum requirements to use IRB approach, 2022. URL https://www.bis.org/committees/bcbs/basel-framework/standard/cre/36/inforce/2023-01-01/published/2022-12-08. Version effective January 1, 2023, published December 8, 2022. Paragraphs 36.10, 36.117, 36.122–36.126 and 36.133–36.140. Consulted October 9, 2026.

Basel Committee on Banking Supervision. Basel Framework CRE20: Standardised approach—individual exposures, 2025. URL https://www.bis.org/committees/bcbs/basel-framework/standard/cre/20/inforce/2023-01-01/published/2025-06-10. Version effective January 1, 2023, updated June 10, 2025. Paragraphs 20.2, 20.4, 20.63, 20.65 and 20.70. Consulted October 7, 2026.

Basel Committee on Banking Supervision. BCBS charter, n.d. URL https://www.bis.org/committees/bcbs/charter. Sections 3 and 5: legal status and members' responsibilities. Consulted October 7, 2026.

Consumer Financial Protection Bureau. Official interpretation of 1002.9: Notifications, n.d.a. URL https://www.consumerfinance.gov/rules-policy/regulations/1002/Interp-9. Paragraph 9(b)(2), consulted September 24, 2026.

Consumer Financial Protection Bureau. 12 cfr 1002.9: Notifications, n.d.b. URL https://www.consumerfinance.gov/rules-policy/regulations/1002/9/. Current online text consulted September 24, 2026.

European Parliament and Council of the European Union. Regulation (EU) 2022/2554 on digital operational resilience for the financial sector. Official Journal of the European Union, L 333, December 27, 2022, pp. 1–79, 2022. URL https://eur-lex.europa.eu/eli/reg/2022/2554/oj. Articles 2, 5–6 and 64. Applies from January 17, 2025. Consulted October 7, 2026.

FFIEC and Consumer Financial Protection Bureau. Public hmda: Lar data fields, n.d. URL https://ffiec.cfpb.gov/documentation/publications/loan-level-datasets/lar-data-fields. Official public data dictionary, consulted September 24, 2026.

Financial Stability Board. The financial stability implications of artificial intelligence. Technical report, Financial Stability Board, November 2024. URL https://www.fsb.org/uploads/P14112024.pdf. Published November 14, 2024. Consulted October 9, 2026.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://arxiv.org/abs/1705.07874.

Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1135–1144, 2016. doi: 10.1145/2939672.2939778.

UCI Machine Learning Repository. South german credit, 2019. URL https://archive.ics.uci.edu/dataset/522/south+german+credit. Dataset; CC BY 4.0. Retrieved September 24, 2026.